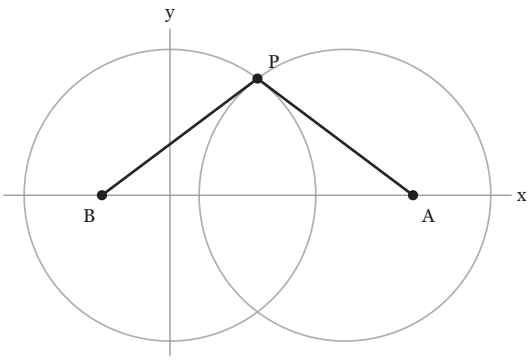
2026 Predictive Paper · Paper 1

Question 1

The circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$ intersect at P, which lies above the x-axis.

The tangents to the two circles at P meet the x-axis at A and B.

What is the area of triangle PAB?



The two tangents meet at P. Use the equations to establish exact coordinates.

A $\frac{32}{3}$

B 16

C $\frac{64}{3}$

D $\frac{100}{3}$

E 25

2026 Predictive Paper · Paper 1

Question 2

For which real values of a does the equation

$\sqrt{x+a} = x - 1$

have exactly one real solution, with that solution strictly between 1 and 3?

A $-\frac{5}{4} < a \leq -1$

B $-1 < a < 1$

C $a = -\frac{5}{4}$ or $-1 < a < 1$

D $-\frac{5}{4} \leq a < 1$

E $a = -\frac{5}{4}$ or $-1 \leq a \leq 1$

2026 Predictive Paper · Paper 1

Question 3

A sequence is defined by $u_1 = 3$ and

$u_{n+1} = \frac{3u_n - 1}{u_n + 1}$ for every positive integer n.

What is $u_1 u_2 \cdots u_{20}$?

A 21

B 420

C 462

D 210

E 231

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Question 4

For $0 \leq x \leq 2\pi$, consider the equation

$$\frac{\sin^2 x}{1 - \cos x} = 2 \sin x$$

What is the sum of $\cos x$ over all its distinct real solutions in this interval?

A $-\frac{2}{5}$

B $\frac{8}{5}$

C $\frac{3}{5}$

D 0

E -1

2026 Predictive Paper · Paper 1

Question 5

A continuous function f, defined for all real x, satisfies

$f(x) - f(-x) = 2x^3 - 6x$

What is $\int_{-2}^{2} x f(x)\,dx$?

A $\frac{16}{5}$

B $\frac{32}{5}$

C $-\frac{32}{5}$

D $-\frac{16}{5}$

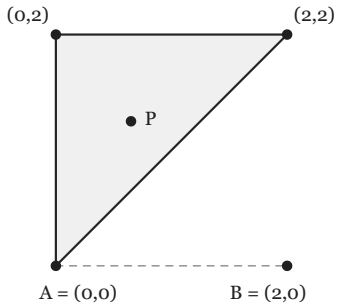
E 0

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Question 6

Let $A = (0, 0)$ and $B = (2, 0)$. A point P is chosen uniformly by area from the interior of the triangle with vertices $(0, 0)$, $(0, 2)$ and $(2, 2)$.

What is the probability that angle APB is greater than 90°?



P is chosen uniformly by area from the shaded triangle.

A $\frac{\pi-2}{4}$

B $\frac{4-\pi}{8}$

C $\frac{\pi}{8}$

D $\frac{\pi-2}{8}$

E $\frac{\pi}{4}$

2026 Predictive Paper · Paper 1

Question 7

The positive integer n is such that the coefficient of x^4 in

$(1-x)^n(1+x)^{n+2}$

is 27. What is the coefficient of x^6 in the same expression?

A -48

B 120

C -84

D -120

E 48

2026 Predictive Paper · Paper 1

Question 8

What is the total length of the locus of points (x, y) satisfying

$\log_2(5 - y) + \log_2(y + 1) = 2\log_2(|x| - 2)$

A 9π

B 12π

C 3π

D 6π

E 4π

2026 Predictive Paper · Paper 1

Question 9

In triangle ABC, the vertices are $A = (0, 0)$, $B = (6, 0)$ and $C = (0, 8)$.

At time $t = 0$, P starts at A and moves along AB at 3 units per second. At the same time, Q starts at B and moves along BC at 5 units per second. Each stops on reaching the end of its segment.

What is the least distance between P and Q during $0 \leq t \leq 2$?

A 4

B 3

C $\frac{6\sqrt{13}}{13}$

D $\frac{12}{5}$

E $\frac{12\sqrt{13}}{13}$

2026 Predictive Paper · Paper 1

Question 10

A histogram represents exactly 20 measurements. The class intervals and frequency densities are:

Class interval	$[0, 1)$	$[1, 3)$	$[3, 6]$
Frequency density	6	4	2

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

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Question 11

Two distinct tangents to the parabola $y = x^2$ pass through the point $(3, -3)$.

What is the area of the triangle formed by these two tangents and the y-axis?

A $36\sqrt{3}$

B $24\sqrt{3}$

C 54

D $18\sqrt{3}$

E $72\sqrt{3}$

2026 Predictive Paper · Paper 1

Question 12

Let $f(x) = |x^2 - 4x + 3|$. For which positive real values of k does

$f(f(x)) = k$

have exactly eight distinct real solutions?

A $0 < k < 1$

B $k > 3$

C $k = 1$

D $k = 3$

E $1 < k < 3$

2026 Predictive Paper · Paper 1

Question 13

How many unordered pairs of distinct integers from $1, 2, \ldots, 30$ have a product that is a perfect square?

A 20

B 24

C 18

D 19

E 16

2026 Predictive Paper · Paper 1

Question 14

For $0 < x < \pi/2$, the infinite series

$\sin^2 x + \sin^4 x + \sin^6 x + \cdots$

has sum $3\cos^2 x$. What is $\tan^2 x$?

A $\frac{7-\sqrt{13}}{6}$

B $\frac{3}{2}$

C $\frac{\sqrt{13}-1}{2}$

D 3

E $\frac{\sqrt{13}+1}{2}$

2026 Predictive Paper · Paper 1

Question 15

A sensor reports height $h(x) = 2x$ for $0 \le x \le 1$, and $h(x) = 6 - x$ for $1 < x \le 2$. It is to be replaced by the constant reading c that minimises $\int_0^2 |h(x) - c|\,dx$. Which option gives both the minimum error and the full interval of minimising values of c?

A $3, \quad 2 \le c \le 4$

B $\dfrac{7}{2}, \quad 2 \le c \le 4$

C $\dfrac{7}{2}, \quad c = 3$

D $4, \quad 1 \le c \le 4$

E $4, \quad c = 3$

2026 Predictive Paper · Paper 1

Question 16

The equation

$3^{2x} - a3^x + 9 = 0$

has distinct real roots u and v satisfying $u^2 + v^2 = 10$. What is a?

A $\frac{28}{3}$

B $\frac{244}{9}$

C 26

D $\frac{82}{3}$

E 28

2026 Predictive Paper · Paper 1

Question 17

Triangle ABC has $AB = AC = 5$ and $BC = 6$, and is inscribed in a circle.

The tangents to the circle at B and C meet at T, on the opposite side of BC from A.

What is AT?

A 20

B $\frac{125}{7}$

C $\frac{50}{7}$

D $\frac{100}{7}$

E $\frac{72}{7}$

2026 Predictive Paper · Paper 1

Question 18

Let $f(x) = x^2 - 4x$ and suppose $-4 < a < 0$.

The curves $y = f(|x|)$ and $y = f(x - a)$ enclose a region of area 3.

What is the sum of the squares of all possible values of a?

A 6

B 16

C 12

D 8

E 10

2026 Predictive Paper · Paper 1

Question 19

A monic polynomial P of degree 4 has remainder $2x + 1$ when divided by $x^2 - 1$, and remainder $x + 7$ when divided by $x^2 - 4$.

What is the least value of $P(x) + P(-x)$ for real x?

A $\frac{3}{4}$

B 6

C $\frac{3}{2}$

D 3

E $\frac{9}{2}$

2026 Predictive Paper · Paper 1

Question 20

For which real values of a does

$4^x - 2a\,2^x + a + 3 = 0$

have two distinct real solutions, both strictly between 0 and 2?

A $\frac{1+\sqrt{13}}{2} < a < \frac{19}{7}$

B $\frac{1+\sqrt{13}}{2} \leq a < \frac{19}{7}$

C $1 < a < \frac{19}{7}$

D $\frac{1+\sqrt{13}}{2} < a \leq \frac{19}{7}$

E $\frac{1+\sqrt{13}}{2} < a < 4$

2026 Predictive Paper · Paper 2

Question 1

A function f, defined for every real number, has exactly four distinct real zeros.

The equation $f(x + 1/x) = 0$, with $x \neq 0$, has exactly five distinct real solutions.

How many of the zeros of f must lie strictly between -2 and 2?

A 2

B 3

C 0

D 4

E 1

2026 Predictive Paper · Paper 2

Question 2

A student attempts to prove that, for all real a, b,

$$\sqrt{a^2+1}+\sqrt{b^2+1} \geq \sqrt{(a+b)^2+4}$$

The steps are:

I. $(a-b)^2 \geq 0$.

II. Therefore $(a^2+1)(b^2+1) \geq (ab+1)^2$.

III. Therefore $\sqrt{(a^2+1)(b^2+1)} \geq ab+1$.

IV. Therefore $(\sqrt{a^2+1}+\sqrt{b^2+1})^2 \geq (a+b)^2+4$.

V. Taking nonnegative square roots gives the claimed inequality.

Which is the first unjustified step, or is the proof valid?

A V

B III

C II

D IV

E The proof is valid.

2026 Predictive Paper · Paper 2

Question 3

Let (u_n) be a real sequence indexed by the positive integers. Consider the statement:

"For every real $a \in [1, 2]$, there is a positive integer N such that, for every integer $n \geq N$, either $u_n < a$ or $u_n > 2a$."

Which is its negation?

A There is a real $a \in [1, 2]$ and a positive integer N such that $a \leq u_n \leq 2a$ for every integer $n \geq N$.

B There is a real $a \in [1, 2]$ such that, for every positive integer N, there is an integer $n \geq N$ with $a \leq u_n \leq 2a$.

C There is a real $a \in [1, 2]$ such that, for every positive integer N, there is an integer $n \geq N$ with $a < u_n < 2a$.

D For every real $a \in [1, 2]$ and every positive integer N, there is an integer $n \geq N$ with $a \leq u_n \leq 2a$.

E For every real $a \in [1, 2]$, there is a positive integer N such that $a \leq u_n \leq 2a$ for every integer $n \geq N$.

2026 Predictive Paper · Paper 2

Question 4

Consider the claim:

"Every real polynomial f with $f(0) < 0$, $f(2) > 0$ and $f'(0) > 0$ has exactly one zero in $(0, 2)$."

Which of the following polynomials is a counterexample?

A $f(x) = -4x^3 + 12x^2 - 11x + 3$

B $f(x) = x^3 + x - 1$

C $f(x) = 4x^3 - 12x^2 + 11x - 3$

D $f(x) = x^3 - x - 1$

E $f(x) = 4x^3 - 12x^2 + 8x$

2026 Predictive Paper · Paper 2

Question 5

Let n be a positive integer. Which condition is necessary and sufficient for n^3 to be divisible by 72 while n^2 is not divisible by 72?

A $n = 6k + 3$ for some integer $k \geq 0$.

B n is divisible by 6.

C $n = 12k + 6$ for some integer $k \geq 0$.

D n is divisible by 12.

E $n = 24k + 12$ for some integer $k \geq 0$.

2026 Predictive Paper · Paper 2

Question 6

Let a, b, c be positive real numbers. Which condition is sufficient but not necessary for a, b, c to be the side lengths of a nondegenerate triangle?

A $1/a, 1/b, 1/c$ are the side lengths of a nondegenerate triangle.

B $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are the side lengths of a nondegenerate triangle.

C a^2, b^2, c^2 are the side lengths of a nondegenerate triangle.

D $a+b, b+c, c+a$ are the side lengths of a nondegenerate triangle.

E The largest of a, b, c is less than the sum of the other two.

2026 Predictive Paper · Paper 2

Question 7

A polynomial f satisfies $0 \le f(x) \le 1$ for $0 \le x \le 1$, and $\int_0^1 f(x)\,dx = 1/2$.

Consider the statements:

I. $\int_0^1 [f(x)]^2\,dx \le 1/2$.

II. $\int_0^1 [f(x)]^2\,dx \ge 1/4$.

III. $\int_0^1 x f(x)\,dx \ge 1/4$.

Which must be true?

A II and III only

B I only

C I and II only

D II only

E I, II and III

2026 Predictive Paper · Paper 2

Question 8

For which real values of a does the polynomial

$f(x) = x^5 - 5ax^3 + 5x$

satisfy $f'(x) > 0$ for every real x?

A $|a| < \frac{2}{3}$

B $a \leq \frac{2}{3}$

C $a > -\frac{2}{3}$

D $a < \frac{2}{3}$

E $|a| \leq \frac{2}{3}$

2026 Predictive Paper · Paper 2

Question 9

A continuous function f is defined on the real line. For one fixed positive real number a, it is known that $\int_0^a f(t)\,dt = a^2$.

A student writes:

I. $\int_0^a f(t)\,dt = a^2$.

II. Define $F(x) = \int_0^x f(t)\,dt$; then $F(a) = a^2$.

III. Differentiating this equality with respect to a gives $F'(a) = 2a$.

IV. Therefore $f(a) = 2a$.

You may use the fact that $F'(x) = f(x)$. Which is the first unjustified step, or is the argument valid?

A II

B IV

C The argument is valid.

D III

E I

2026 Predictive Paper · Paper 2

Question 10

Two histograms, A and B, each describe eight real observations. Their class intervals and frequency densities are:

Class interval	$[0, 2)$	$[2, 4)$	$[4, 8]$
Density in A	1	1	1
Density in B	2	1	$1/2$

The median of eight observations is the mean of the fourth and fifth observations in increasing order.

Consider the statements:

I. The mean in A is greater than the mean in B.

II. The median in A is greater than the median in B.

III. The maximum in A is at least the maximum in B.

Which must be true?

A III only

B II only

C I and II only

D I only

E II and III only

2026 Predictive Paper · Paper 2

Question 11

A real polynomial f of degree exactly 4 has exactly three distinct stationary points, at $x = a, b, c$ where $a < b < c$.

It satisfies $f(a) = f(c) = 0$ and $f(b) > 0$.

Consider the statements:

I. $b = (a + c)/2$.

II. $f(x) \geq 0$ for every real x.

III. For every real $d > 0$, the equation $f(x) = d$ has exactly four distinct real solutions.

Which must be true?

A II only

B I and III only

C I only

D I, II and III

E I and II only

2026 Predictive Paper · Paper 2

Question 12

For a real parameter a, consider:

P: For every $x \in [0, 1]$, there is a $y \in [0, 1]$ such that $(x - y)^2 < a$.

Q: There is a $y \in [0, 1]$ such that, for every $x \in [0, 1]$, $(x - y)^2 < a$.

For which values of a is P true and Q false?

A $a \geq \frac{1}{4}$

B $a > \frac{1}{4}$

C $0 < a < \frac{1}{4}$

D $0 < a \leq \frac{1}{4}$

E $a \leq 0$

2026 Predictive Paper · Paper 2

Question 13

Suppose $a > 0$, $a \neq 1$, $b > 0$, $c > 0$ and

$\log_a b > \log_a c > 0$

Consider the statements:

I. $(a-1)(b-c) > 0$.

II. $0 < \log_b c < 1$.

III. $b + c > 2$.

Which must be true?

A I, II and III

B II and III only

C II only

D I and II only

E I only

2026 Predictive Paper · Paper 2

Question 14

A real polynomial f satisfies

$$\int_0^1 [f(x)]^2\,dx = 2\int_0^1 x f(x)\,dx - \frac{1}{3}$$

What is $f(-2)$?

A -1

B It cannot be determined.

C 0

D -2

E 2

2026 Predictive Paper · Paper 2

Question 15

For real a, b, c, consider

$$x^2 + y^2 + 2ax + 2by + c = 0$$

Which condition is necessary and sufficient for this equation to describe a nondegenerate circle whose entire circumference lies in the open first quadrant $x > 0$, $y > 0$?

A $a < 0$, $b < 0$, and $\max(a^2, b^2) < c < a^2 + b^2$

B $a < 0$, $b < 0$, and $\max(a^2, b^2) \le c < a^2 + b^2$

C $a < 0$, $b < 0$, and $0 < c < a^2 + b^2$

D $a < 0$, $b < 0$, and $c > a^2 + b^2$

E $a > 0$, $b > 0$, and $\max(a^2, b^2) < c < a^2 + b^2$

2026 Predictive Paper · Paper 2

Question 16

A geometric progression has first term $a > 0$ and real common ratio r.

Both infinite series below converge, and

$\sum_{n=1}^{\infty} ar^{n-1} = 2$, while $\sum_{n=1}^{\infty} (ar^{n-1})^2 = 12$.

What is $\sum_{n=1}^{\infty} |ar^{n-1}|$?

A 4

B 6

C 2

D 12

E 8

2026 Predictive Paper · Paper 2

Question 17

What is the least integer k such that every selection of k distinct integers from $1, 2, \ldots, 48$ must contain three numbers $a < b < c$ for which a divides b and b divides c?

A 33

B 37

C 36

D 35

E 25

2026 Predictive Paper · Paper 2

Question 18

A nonempty finite set S consists of real numbers other than 0 and 1.

For every $x \in S$, both $1 - x$ and $1/x$ also belong to S.

What is the least possible number of elements of S?

A 3

B 5

C 1

D 4

E 2

2026 Predictive Paper · Paper 2

Question 19

A continuous function f is defined on the real line and satisfies $f(x+1) = -f(x)$ for every real x.

Consider the statements:

I. $\int_{a}^{a+2} f(x)\,dx = 0$ for every real a.

II. Every open interval $(a, a+1)$ contains a zero of f.

III. $[f(x+1)]^2 = [f(x)]^2$ for every real x.

Which must be true?

A I and II only

B I, II and III

C II only

D I only

E I and III only

2026 Predictive Paper · Paper 2

Question 20

For real parameters p, q, which condition is necessary and sufficient for

$$x^4 - px^3 + qx^2 - px + 1 = 0$$

to have four distinct positive real roots?

A $p > 4$ and $2p - 2 < q < \frac{p^2}{4} + 2$

B $p > 0$ and $2p - 2 < q < \frac{p^2}{4} + 2$

C $p > 4$ and $2p - 2 \leq q < \frac{p^2}{4} + 2$

D $p > 4$ and $2p + 2 < q < \frac{p^2}{4} + 2$

E $p > 4$ and $2p - 2 < q \leq \frac{p^2}{4} + 2$